"The IoT is great, but comes with security challenges for developers"

The time has come for embedded Linux to rule. From consumer electronics (like set-top boxes and smart TVs), in-vehicle infotainment (IVI), networking equipment (wireless routers) and industrial automation, to spacecraft flight software and medical instruments, embedded Linux offers endless possibilities. Android has pushed the frontiers of this domain to an entirely new level with devices getting launched every other day. But though the use of embedded Linux is growing, there are challenges that it faces. The biggest one, perhaps, is the need for stronger security capabilities.

Open Source For You spoke to ***Alok Mehrotra, country manager (India) and Rajesh Lalwani, account manager, Wind River*** about how the company's 'security profile' helps combat the everyday threats that are a normal offshoot of the Internet of Things (IoT). Excerpts:

Q Do brief us about Wind River's presence in the open source domain.

I'd like to share a little bit of history with you. VxWorks is our real time operating system (RTOS). You will find that pretty much any mission-critical embedded design you can think of will be on an RTOS, and that RTOS will typically be Wind River. To give you some perspective, today, we have over one and a half billion devices running on Wind River as the operating system, on which you go ahead and build your applications.

Commercial embedded Linux continues to gain traction across the board as the aerospace, defence, industrial, networking and automotive industries see how open source encourages rapid innovation at far lower costs. But navigating the open source ecosystem is not an easy task, and Linux is different from the real time operating systems, software development tools and test frameworks that play a critical role in developing and deploying embedded devices. Thanks to the thousands of developers who strive to make it better, Linux constantly evolves and expands over time.

Alok Mehrotra, country manager (India), Wind River

Currently, we are experiencing the Internet of Things (IoT) phenomenon, where everything is connected to everything else. It's an amazing concept. However, the major challenge is how to secure the network using open source technology, which is available to everybody. In a nutshell, the IoT is great, but it comes with a few security challenges for developers.

Wind River has come up with a security profile on top of Linux, which secures the operating system besides being Evaluation Assurance Levels-4 (EAL-4) certified. It's a commercial off-the-shelf (COTS) product, built to align with the Common Criteria for Operating System Protection Profile. In this, we've hardened the kernel, enhanced the user space and have given the entire control of the user space to the super user. The security-focused kernel includes features like grsecurity, PaX and enhanced Address Space Layout Randomisation (ASLR), among others.

Q What are the complex issues that developers face with the Internet of Things (IoT)?

Everybody has their own definition for the Internet of Things. Connectivity, manageability and security are important aspects. To me, the bigger question from the IoT perspective is not just how I connect to an aggregator, but how does this whole thing happen with the edge of the cloud.

The IoT has various applications such as smart cities, for instance, where the concept can be widely implemented with connected cars. Industrial automation is another application, where everything you need is on the Internet, so you can regularly update it. In the aforementioned cases, you need various protocols since you want everything to be secure, and these must be connected on the cloud as well. We need security at two levels. One, so that nobody can attack your system, and two, so that nobody can install their own application on top of it. In case they want to install a particular application, they need to have some authorised access to it.